INCH-POUND
MIL-DTL-24643/8D
22 August 2002
SUPERSEDING
MIL-C-24643/8C
14 March 1994

DETAIL SPECIFICATION SHEET

CORD, ELECTRICAL, 300 VOLTS, TYPE LSMMOP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor - AWG size No. 24, class O stranding, uncoated.
Second -	Separator may be used at manufacturer's option where required to provide free stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation. Minimum average wall thickness 0.020 inch. Standard identification code shall be applied by method 3.
Fourth -	Five conductors cabled together with a lay not greater than 2 inches. A tie cord shall be laid straight as the central core. Fillers shall be used as necessary, to obtain a firm well-rounded assembly. Cabling sequence shall be consecutive.
Fifth -	Binder of polyester tape or braid.
Sixth -	Cross-linked polyolefin jacket.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/8	Type and size	Number of conductors	Insulated conductor minimum (inch)	Overall maximum (inch)
-01UN	LSMMOP-5	5	0.065	0.305

EXAMINATION AND TESTS:

Basic electrical: Requirements:

- Conductor resistance - ohms/1000 feet at 25°C, maximum 27.96
- Voltage withstand - volts, root mean square, minimum 1000
- Insulation resistance - megohms -1000 feet, minimum 100
- Conductor continuity ..... No failure

Group A:

- Visual and dimensional ..... No failure

Group B:

- Cross-linked proof test (percent, maximum)
  - Insulation 50
- Jacket (When tested at 200°C) 50
- Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)

- Insulation
  - Ethylene propylene rubber
    - Tensile strength - lb/in^2, minimum 700
    - Elongation - percent, minimum 250
  - Cross-linked polyethylene
    - Tensile strength - lb/in^2, minimum 1800
    - Elongation - percent, minimum 250
  - Jacket (cord)
    - Tensile strength - lb/in^2, minimum 1300
    - Elongation - percent, minimum 160
  - Twisting and bending endurance -
    - At 75°C, cycles, minimum 6000
    - At minus 30 ± 2°C, cycles, minimum 2000

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Group C:
Physicals (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Permanence of printing (jacket) - cycles, minimum 125
Heat distortion - percent of unaged, maximum 30

Group D:
Flame propagation (cable..... No failure

QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ± 5°C) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Halogen content - percent, maximum
Jacket 0.2
Fillers 0.2
Immersion (jacket)
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Smoke index, maximum
Jacket 25
Fillers 45
Insulation 45
Toxicity index, maximum

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Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:
All sizes 500 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-006)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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